Burhanudin

burhanudin.at.work.01@gmail.com | linkedin.com/in/burhanudin-898505220 | burhanudin02.github.io

Biomedical Engineering graduate focused on medical technology and AI-driven control systems. Experienced in reinforcement learning for surgical robot simulation, ROS and computer vision for autonomous systems, and brain signal acquisition using ECVT technology. Demonstrated hands-on expertise in Python-based ML implementation, robotic control, and experimental neuroimaging research within academic and research environments.

Education

Universitas Gadjah Mada — Yogyakarta, Indonesia
B. Eng. in Biomedical Engineering — *2021 – 2026*

Experiences

Universitas Gadjah Mada — Yogyakarta, Indonesia
Assistant Programmer — ROS2 (Autonomous Excavator ERIC Project) — *Apr 2024 – Nov 2024*

- Developed ROS2 nodes for perception and control integration in an autonomous excavator system.
- Implemented computer vision pipelines for object detection and environmental awareness.

C-Tech Labs Edwar Technology — Tangerang, Indonesia
Research Intern — Neuroimaging & Brain Signal Analysis — *Jan 2024 – Feb 2024*

- Performed frequency sweep characterization of a brain signal acquisition module using a function generator and digital oscilloscope, and developed Python-based analysis to compare injected carrier signals with captured neural responses to identify optimal operating frequency.
- Assisted in operating the 4D Brain Scan system for patient assessments.
- Gained practical experience in neuroimaging technology and its applications in both clinical and research settings.

GAMAFORCE UGM — Yogyakarta, Indonesia
Vision & Control Programmer — *Dec 2022 – Dec 2024*

- Developed vision-based guidance and control algorithms for autonomous aerial systems.
- Integrated computer vision modules with control systems for target tracking and navigation tasks.
- Collaborated within multidisciplinary engineering teams for system testing and field validation.

Jago Robotika — Yogyakarta, Indonesia
Student Tutor — *Nov 2024 – Dec 2025*

- Mentored wide range of students from kindergarten to junior high school level in robotics fundamentals, programming, and system integration concepts.
- Facilitated hands-on robotics sessions guiding students in developing their own Arduino-based projects integrating sensor inputs and actuator outputs.

Projects

Bachelor Thesis Project – AI-Based Surgical Robot Control | *Python, Reinforcement Learning*

- Developed a reinforcement learning framework for autonomous surgical tool manipulation using the da Vinci Research Kit simulator.
- Engineered reward functions and exploration mechanisms to improve convergence and task success in needle-picking scenarios.
- Conducted comparative analysis of training configurations to identify the most stable and efficient learning strategy.

Capstone Project – BISINDO Sign Language Translator App | *Dataset Acquisition*

- Led end-to-end development of a BISINDO sign language translation system, overseeing data acquisition, preprocessing, and model training workflows.
- Worked with certified BISINDO practitioners to capture structured sign language gesture dataset.
- Performed dataset annotation, preprocessing, and augmentation to enhance robustness of the model.
- Extracted time series data of hand landmarks position from the dataset for a LSTM model training data.

Senior Project – Analog Line Follower Robot | *Sensors, Electronics*

- Developed a mobile robot to follow a line track by utilizing infrared sensors and operational amplifiers.
- Redesigned the electrical circuit and manufactured the PCB for proper component placement.

Visual Guidance System for UAV | *Computer Vision, Control Systems*

- Developed a visual based object tracking system to guide UAV in performing a payload picking and precision dropping mission.
- Interpreted visual feedback and fine-tuned controller parameters to achieve a stable movement behavior.

Personal Project – Robot Simulation | *MuJoCo, Reinforcement Learning*, Onshape

- Trained a reinforcement learning agent to navigate the end-effector of KUKA LBR IIWA robot.
- Designed gripper mechanism as additional attachment module for the robot.

Personal Project – Computer Aided Design | *Onshape*

- Designed a 3D model assembly for medium sized DIY CNC milling machine.
- Researched the locally available parts and components for the design.

Personal Project – Self-hosted AI Chat Bot | *Ollama, Docker*

- Set up a self-hosted AI chat interface using Open-WebUI, Ollama, and Docker stack.
- Applied prompt engineering to optimize the chatbot's behavior in chat response and tool calling workflows.

Publications

15th Asian Control Conference (ASCC) Bali, Indonesia
Analyzing Reward Design Effects in PPO for Surgical Needle Picking in SurRoL. *Jun 2026*

Technical Skills

Embedded & Control: C/C++, ESP32, Arduino, SBC, Linux
Engineering Tools: LTSpice, Arduino IDE, ROS, Gazebo, MuJoCo, FreeCAD, Onshape
Machine Learning: Python
Languages: Bahasa Indonesia (Native), English (C1), Arabic (beginner), Mandarin (beginner)